APPENDIX 18:

Review & Recommendation Report

Project Title: _____

Fair Division: [ ] Life Science [ ] Physical Science [ ] Robotics and Intelligent Machines
[ ] Mathematics and Computational Science
Category: [ ] Individual [ ] Team

Instruction: Please put a check [] in the appropriate column and if necessary, write recommendations on the space provided.

PART 1: REQUIRED FORMS FOR ALL RESEARCHES	COMPLETE	INCOMPLETE	RECOMMENDATIONS
1. Checklist for Adult Sponsor (1). Is it accomplished and signed?			
2. Student Checklist 1A. Is it complete and signed?			
If answer to item 5 is **YES**, must also have Form 7 (See Part II, item 13 below)			
If answer to item 7 is **Research Institution** or **Other**, must also have Form IC (See Part II, item 6 below)			
3. Research Plan. (Attachment to item 2 above). Does it include the following:			
A. Rationale. Does it Include a synopsis of background information that supports the research problem and explains why the research is important scientifically? If applicable, does it explain the societal impact of the research?			
B. HYPOTHESIS(ES), RESEARCH QUESTION(S), ENGINEERING GOAL(S), EXPECTED OUTCOMES. Is this based on RATIONALE?			
C. RESEARCH METHODS AND CONCLUSIONS. a. Procedures. i. Does it show all procedures and experimental designs, including methods for data collection? ii. There should be NO inclusion of work of mentor or others. iii. Parameters should NOT be too strict to allow for possible changes.			

b. Risk and Safety. Does it identify all potential risks and safety precautions needed?			
c. Data Analysis. i. Does it describe all procedures for data analysis? ii. Parameters should NOT be too strict to allow for possible changes.			
D. BIBLIOGRAPHY. Does it have at least 5 major references? If using vertebrate animals, include 1 reference on animal care. (American Psychological Association)			
Note: Items 3.E-H are needed ONLY for researches on HUMAN PARTICIPANTS, VERTEBRATE ANIMAL, POTENTIALLY HAZARDOUS BIOLOGICAL AGENTS (see attached Research Plan/Project Summary Instructions)			
E. HUMAN PARTICIPANTS RESEARCH. Does it provide for the following? a. Description b. Recruitment c. Methods d. Risk Assessment e. Protection of Privacy f. Informed Consent Process			
F. VERTEBRATE ANIMAL RESEARCH. Does it provide for the following? a. Potential ALTERNATIVES to vertebrate animal use b. Potential impact or contribution of research c. Detailed procedures d. Detail animal numbers, strain, sex , age, source, etc. e. Describe housing and oversight of daily care f. Disposition of animals at study termination			
G. POTENTIALLY HAZARDOUS BIOLOGICAL AGENTS RESEARCH. Does it provide for the following? a. Biosafety Level (BSL) Assessment and Determination b. Source of agent, specific cell line. c. Safety precautions d. Methods of disposal			
4. Approval Form 1B (for ALL students)			
5. Abstract			

VERY IMPORTANT: See Part II, Risk Assessment (3) for

a. Studies involving protists, archaea and similar microorganisms.
b. Research using manure for composting, fuel production, or other non-culturing experiments.
c. Commercially-available color change coliform water test kits. These kits must remain sealed and must be properly disposed.
d. Studies involving decomposition of vertebrate organisms (such as in forensic projects).
e. Studies with microbial fuel cells.

PART 2: ADDITIONAL REQUIRED FORMS	COMPLETE	INCOMPLETE	RECOMMENDATIONS
6. Regulated Research Institutional or Industrial Setting Form (1C). Must be completed AFTER experimentation by the adult supervising the student research conducted in a regulated research institution or any work site aside from home, school or field. **Is it properly accomplished and signed by the DESIGNATED SUPERVISING ADULT?**			
7. Qualified Scientist Form (2) - for researches with human participants, vertebrate animals, potentially hazardous biological agents, Drug Enforcement Administration (DEA)-controlled substances; completed and signed BEFORE start of experimentation. **Is it properly accomplished and signed by the QUALIFIED SCIENTIST?**			
8. Risk Assessment Form (3) – for researches using hazardous chemicals, activities or devices and microorganisms exempt from pre-approval. Must be completed BEFORE experimentation. **Is it properly accomplished and signed by DESIGNATED SUPERVISING ADULT OR QUALIFIED SCIENTIST (when applicable)?**			
9. Human Participants Form (4) – for researches involving human participants not at a Regulated Research Institution. **Did the DESIGNATED ADULT SUPERVISOR/INSTITUTION approve the research BEFORE experimentation?**			

10. Vertebrate Animal Form (5A) – for researches involving vertebrate animals that is conducted in a school/home/ field research site. **A. Is it properly accomplished, approved and signed by SRC BEFORE experimentation?** **B. Is it properly accomplished, approved and signed by DESIGNATED VETERINARIAN BEFORE experimentation?** **C. Is it properly accomplished, approved and signed by DESIGNATED SUPERVISOR OR QUALIFIED SCIENTIST (as applicable) BEFORE experimentation?**			
11. Vertebrate Animal Form (5B) – for researches involving vertebrate animals that is conducted at a Regulated Research Institution. **A. Does it have IACUC approval BEFORE experimentation?** **B. Is it properly accomplished, approved and signed by a QUALIFIED SCIENTIST/ PRINCIPAL INVESTIGATOR?**			
12. Potentially Hazardous Biological Agents Risk Assessment Form (6A) – for researches involving microogranisms, rDNA, fresh/ frozen tissue (including primary cell lines, human and other primate established cell lines and tissue cultures), blood, blood products and body fluids. **A. Does it have SRC/IACUC/ Institutional Biosafety Committee (IBC) approval BEFORE experimentation?** **B. Is it properly accomplished, approved and signed by a QUALIFIED or DESIGNATED SUPERVISOR BEFORE experimentation?** **C. Is it properly accomplished, approved and signed by the SRC BEFORE experimentation?** **D. Human Vertebrate Animal Tissue Form (6B) –** for researches involving fresh/ frozen tissue (including primary cell lines, human and other primate established cell lines			

and tissue cultures), blood, blood products and body fluids. If research involves living organisms, ensure that the proper human or animal forms are completed. All researches using any tissue listed above must also complete Form 6A. Is it properly accomplished, approved and signed by a QUALIFIED or DESIGNATED SUPERVISOR BEFORE experimentation?			
13. Continuation/Research Progression Projects Form (7) – for researches that are a continuation/progression in the same field of study as a previous research. **A. 13.This form MUST be accompanied by the PREVIOUS YEAR'S ABSTRACT and RESEARCH PLAN** **B. Is it properly accomplished, approved and signed by the student/s?**			

PART 3: RESEARCH PAPER (See attached IMRAD Format)	COMPLETE	INCOMPLETE	RECOMMENDATIONS
1. COVER PAGE A. Is the research title present? B. Part 3 1 B: Is/Are the name/s of the student proponent/s present? C. Is/Are the appropriate persons credited? (The Research adviser and Research Consultants, if applicable MUST be present)			
2. INTRODUCTION Does it outline the research question and its significance within the topic discussed, making its relevance clear to readers in a CONCISE manner?			
3. METHOD Does it clearly and comprehensively provide the reader with a description of the methods used in the research?			
4. RESULTS Does it clearly and comprehensively SHOW the reader what the research came			

up with? This should be the MAIN section of the paper.			
5. DISCUSSION Does this show what the findings in RESULTS mean?			
6. LIMITATIONS ON THE RESEARCH DESIGN AND MATERIAL Does this show knowledge and understanding of research limitations?			
7. CONCLUSION, NOTES, WORKS CITED AND APPENDICES/BIBLIOGRAPHY A. Does the conclusion briefly and clearly analyze what the paper proposed, discussed and concluded? B. Are there (in MLA format) possible Researcher Notes, the research paper's Works Cited, and Appendices?			

PART 4: RESEARCH ABSTRACT (MAX. 250 WORDS)	COMPLETE	INCOMPLETE	RECOMMENDATIONS
1. Does it clearly and concisely state the PURPOSE OF THE RESEARCH?			
2. Does it clearly and concisely state the PROCEDURE/S undertaken in the RESEARCH?			
3. Does it clearly and concisely state the DATA collected from the RESEARCH?			
4. 4.Does it clearly and concisely state the CONCLUSIONS OF THE RESEARCH?			

VERY IMPORTANT: There should be NONE of the following:

a. Acknowledgements of the research institutions and/or mentors with which the student were working
b. Self-promotions and external endorsements
c. Inclusion of work or procedures done by the mentor

PART 5: RESEARCH LOGBOOK	COMPLETE	INCOMPLETE	RECOMMENDATIONS
1. Is the logbook intact and not tampered with? It should NOT be loose-leafed.			
2. Does the START DATE in the logbook match the START DATE in Student Checklist (1A)?			
3. Does the END DATE in the logbook match the END DATE in Student Checklist (1A)?			
4. Are all the entries in the logbook properly dated?			
5. Does the logbook show accurate and detailed notes and findings throughout the course of the research? Does it include data tables, and the like?			
6. Does the logbook show accurate and detailed description of procedures and processes conducted in the course of the research?			
7. Does the logbook show student notes and questions in the course of the research?			

[ ] Qualified [ ] Disqualified Reviewed by: _____ Date: _____

Reason/s for Disqualification: _____